SVD-Compressed Cross Sections in Monte Carlo Neutron Transport

An implementation-led benchmark with a partition-of-unity fix for off-library temperature interpolation

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Abstract

We test whether a truncated SVD of $\log_{10}\sigma(E,T)$ is a defensible alternative to pointwise cross-section tables, and whether a partition-of-unity variant of Ducru's kernel reconstruction [1] closes the accuracy gap at realistic off-library operating temperatures. A pure-Rust transport engine implements four providers — pointwise table, SVD, ACE+WMP, and SVD+WMP hybrid — behind a common interface, validated against OpenMC 0.15.3 to $|\Delta k| \leq 51$ pcm on a 9-nuclide PWR pin cell and on the Godiva fast benchmark. 43 of 47 non-redundant ^{235}U reaction channels are rank-one to machine precision, and rank-5 SVD with discrete-level rank-1 recovers engine-wide k_{eff} within ~ 50 pcm of the in-engine pointwise baseline on both benchmarks at 1.37×–1.90× the CPU throughput. At a realistic off-library operating point (+150 K above every nuclide's library column, fuel 900 → 1050 K, moderator 600 → 750 K), a partition-of-unity 3-point Ducru scheme reproduces k_∞ within 213 pcm ($\sim 3.6\sigma$) of the ACE+WMP industry baseline while using 22% less memory and matching its transport speed — a 10× reduction over a naive 2-point bracketing variant that leaves a 2094 pcm kernel-approximation bias. The 132.9× representation-byte compression ratio of the literature is a layout comparison, not the 1.06× engine-scale reduction we measure; we make that distinction explicit. We flag three honest failures: GPU SVD is 1.3× *slower* than GPU pointwise on PWR due to clad level-sum cost; the WMP+SVD hybrid is 2× slower than CPU SVD due to Faddeeva evaluation cost; and a simplex-projected QP variant of Ducru (clip negative weights, renormalize) degrades U-238 capture reconstruction from 0.84% to 5% L2 error because the projection is optimal in weight space rather than in the Doppler-kernel norm. The kernel-Gram QP (Ducru [1] §4) *is* implemented and wins at rank 3 (0.60% L2) but loses at our production rank 5 (2.06%), so signed 3-point unity remains the default; an analytic Doppler-kernel Gram QP is the outstanding follow-up.

Keywords: Monte Carlo, cross-section compression, singular value decomposition, windowed multipole, Doppler broadening, partition-of-unity interpolation, nuclear data, GPU transport.

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1 Introduction

Continuous-energy Monte Carlo codes such as OpenMC [2], MCNP, and Serpent store nuclear data as pointwise tables. Per reaction channel and per library temperature, a piecewise-linear or log-log interpolation array carries 10^4–10^5 grid points for light nuclides and 10^5–10^6 for dense-resonance actinides. A full-library actinide set at a handful of temperatures exceeds the working-set of contemporary L2/L3 caches, and every collision pays for a binary search plus a log-log blend through a large monotonic array.

Prior work attacks this bottleneck along three axes. Adaptive resampling machine-learned models retain 10–50% of the pointwise data at ≤ 97 pcm criticality error [3, 4]. The Windowed Multipole (WMP) method [5–7] gives an analytical pole–residue representation of the resolved resonance region with on-the-fly Doppler broadening. GPU rewrites reorganise the transport loop to amortise the lookup across warps [8, 9].

This paper asks a narrower question. Can a truncated SVD of the $\log_{10}$-transformed matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$ replace the pointwise table as a drop-in cross-section representation, and if so, at what cost? We implemented three providers (pointwise table, SVD kernel, and SVD+WMP hybrid) inside a single pure-Rust transport engine, plus matching CUDA paths for pointwise and SVD (and a standalone CUDA WMP evaluator), and measured fidelity, throughput, and memory side-by-side.

The answer, reported in full in Sections 6 and 9, is *qualified*: SVD is a defensible drop-in on the thermal pin cell we tested; the hybrid implementation works end-to-end and reproduces k_∞ inside one SEM of the pointwise baseline; the throughput picture is hardware-dependent and favours pointwise tables on our GPU; and the memory picture at engine scale is *not* the 132.9× reduction a representation-byte analysis suggests. We document the trade-offs explicitly and retract the earlier memory headline.

Contributions.

- A singular-spectrum measurement showing 43 of 47 non-redundant ^{235}U reactions are rank-one to machine precision at $T = 294$ K across six library temperatures, with a per-reaction distribution and Pareto (Sec. 4, Fig. 1).

- A rank-k SVD reconstruction kernel for pointwise cross sections, with Ducru temperature interpolation [1]. Rank-6 reaches machine precision at 44 ns/lookup against 91 ns for a binary-search pointwise table (kernel-level 2.0×).

- A PWR pin cell rank sweep (9 nuclides, S(α,β), URR, discrete levels) showing CPU SVD rank-5 matches the in-engine pointwise table on k_∞ to 13–47 pcm on two hardware tiers at 1.37×–1.90× throughput (Sec. 6).

- A matching event-based CUDA solver [8] for both providers, showing GPU pointwise is 1.30× faster than GPU SVD on PWR pin cell, inverting the CPU ordering (Sec. 8).

- A hybrid SVD+WMP engine: pure-Rust Faddeeva reader and evaluator matching OpenMC Python to $\sim 10^{-6}$ relative error, wired into a `HybridSvdWmpXsProvider`. PWR pin cell $k_\infty = 1.32557 \pm 0.00070$, 86 pcm from the CPU-SVD baseline; hybrid throughput 2.06× slower than CPU SVD. In-engine memory 1.06× the full-SVD baseline (Sec. 9).

- A CUDA port of the WMP evaluator, bit-exact against the CPU path ($\leq 5 \cdot 10^{-14}$ relative at double precision), with 7.0× GPU-vs-CPU per-lookup throughput on an RTX 3080 (Sec. 10).

- A tracing and closure of the original 78–127 pcm engine offset from OpenMC 0.15.3 via three corrections (XS-audit exonerating the total-XS assembly; a $\bar{\nu}$ delayed-yield reader fix; correlated-CDF → stochastic-bin sampling for angular and fission distributions, ported to both CPU and GPU paths). Ten-seed k_∞ on all four providers now agrees with OpenMC to $|\Delta| \leq 51$ pcm (CPU table −12, CPU SVD −25, GPU pointwise +51, GPU SVD −8), with a Shannon-entropy source-convergence monitor added as runtime diagnostic (Sec. 6.3).

- A clear separation of two memory measurements: the WMP representation-byte ratio against a hypothetical four-channel pointwise table is 132.9× for our nine-nuclide set, and the measured in-engine reduction from hybrid over full-SVD is 1.06×. The hybrid engine in absolute terms is 4.8–5.1× larger than the pointwise-table baseline, because discrete-level SVD kernels dominate memory and are not covered by the public WMP library. The gap between representation-byte ratios and engine-scale savings is the main cautionary finding of this paper, and we discuss what would need to be true for a WMP-scale reduction to land at engine level (Sec. 9.1, Sec. 9.4).

2 Method

2.1 Pointwise baseline

For each nuclide and library temperature, $\sigma(E_i, T_j)$ lies on a union energy grid $\{E_i\}_{i=0}^{N_E-1}$. For $E_i \le E < E_{i+1}$ the standard log-log lookup is

$$\sigma(E) = \sigma_i \left(\sigma_{i+1}/\sigma_i\right)^f, \qquad f = \frac{\ln(E/E_i)}{\ln(E_{i+1}/E_i)}. \tag{1}$$

We use a log-uniform hash index with 8192 bins (Brown 2014 [10]) to resolve i in $O(1)$, falling back to a bounded binary search in dense bins. Non-positive cross-section values fall back to linear interpolation to avoid log of a negative.

2.2 Truncated SVD

Pre-clamp the matrix to avoid $-\infty$ in the logarithm, $\mathbf{A}_{ij} = \max(\sigma(E_i, T_j), 10^{-30})$, and compute the thin SVD of $\mathbf{L} = \log_{10} \mathbf{A}$:

$$\mathbf{L} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top, \qquad \mathbf{L} \approx \mathbf{U}_k \boldsymbol{\Sigma}_k \mathbf{V}_k^\top. \tag{2}$$

Pre-multiply the basis $\mathbf{B} = \mathbf{U}_k \boldsymbol{\Sigma}_k$ so a reconstruction at grid point i for target coefficient vector $\mathbf{c} \in \mathbb{R}^k$ is one dot product:

$$\log_{10} \tilde{\sigma}(E_i, T) = \sum_{\ell=1}^{k} B_{i\ell}\, c_\ell, \qquad \tilde{\sigma} = 2^{(\log_{10} \tilde{\sigma}) \cdot \log_2 10}. \tag{3}$$

$\mathbf{B}$ stores $N_E k$ doubles row-major so k values for one energy point live in $8k$ contiguous bytes; at $k = 5$ that is one cache line. Between grid points we apply log-log interpolation on $\log_{10} \tilde{\sigma}$ and take one $\exp_2$.

2.3 Temperature interpolation

Library temperatures are typically $\{0, 250, 294, 600, 900, 1200, 2500\}$ K in the ENDF/B-VII.1 HDF5 distribution. For target $T \notin \{T_j\}$ the coefficient vector follows the Ducru kernel method [1]:

$$c_\ell(T) = \sum_{j \in \mathcal{S}} w_j(T)\, V_{j\ell}, \qquad w_j(T) = \frac{\sqrt{T_j T}}{T_j + T} \prod_{i \in \mathcal{S}, i \ne j} \frac{T - T_i}{T + T_i} \cdot \frac{T_j + T_i}{T_j - T_i}, \tag{4}$$

where $\mathcal{S}$ is a subset of library temperatures used for the interpolation. Eq. 4 is exact at library temperatures (w collapses to a one-hot selector when $T = T_j$), $\sim$0.1% error on Doppler-broadened cross sections over 300–3000 K, and L2-optimal for the free Doppler kernel on $\mathcal{S}$.

Choosing the interpolation subset

Using $|\mathcal{S}| = N$ (all library temperatures) is numerically unstable for $N \gtrsim 4$ because the product-of-ratios term in w_j develops near-singular factors when non-adjacent T_i's bracket T with large stride. In our measurements with $N = 7$ the 2500 K column pulls the product toward values $\sim 10^2$, the resulting weights routinely swing by a factor of 10–100 across neighbouring target temperatures, and the log-space reconstruction sees non-physical oscillations that scale linearly with the condition number.

We use the three library temperatures nearest to T. Three is the smallest subset that captures a quadratic correction to the Faddeeva kernel (two-point misses it; see Sec. 7); it is also small enough that the product term has only two factors per weight and the formula is well-conditioned at every operating temperature we tested in the interval $[250, 2500]$ K. Section 2.4 describes how we make the subset weights a partition of unity; this is essential for resonance-dominated systems and is the single change that moves the off-library PWR result from RED to GREEN on the verdict semaphore.

2.4 Partition-of-unity Ducru for off-library T

When Eq. 4 is applied to $V^\top$ columns of a $\log_{10}\sigma$ SVD, the reconstructed $\log_{10}\tilde{\sigma}(E_i, T)$ is a linear combination of the per-library-T log cross sections with coefficients $w_j(T)$. On the *linear* cross section this is a weighted geometric mean:

$$\tilde{\sigma}(E,T) = \prod_{j\in\mathcal{S}} \sigma(E,T_j)^{w_j(T)}. \tag{5}$$

If $\sum_j w_j \neq 1$, the right-hand side of Eq. 5 is scaled uniformly by $\exp[(\sum_j w_j - 1)\cdot\overline{\log\sigma}]$ where the overbar denotes a weighted average of $\log_{10}\sigma(E,T_j)$. For smooth channels this is a barely-visible gain error (far away from resonance peaks, the average $\log\sigma$ is small and the multiplicative shift is close to unity). On narrow resonance peaks it is catastrophic: at the U-238 6.67 eV resonance with a peak cross section of $\sim$4500 b, a 5% gain error moves the peak by $\sim$225 b, which on a PWR pin cell translates to $\sim$2000 pcm on k_∞ (Sec. 7).

Raw Ducru does *not* have $\sum_j w_j = 1$ in general. Eq. 4's weights are designed to reconstruct $\sigma(E,T)$ in the free-Doppler-kernel L2 norm (see [1], §3); partition of unity is not one of the optimization's constraints. Measured on our $N = 3$ bracketing subset at $T = 750$ K with library $(600, 900, 1200)$ K the raw weights $(0.21, 0.30, -0.01)$ sum to 0.50. At $T = 1050$ K with library $(900, 1200, 2500)$ K they sum to 0.51. At endpoints they collapse to $(1, 0, 0)$ / $(0, 1, 0)$ so the total is 1; *between* endpoints the sum drifts by up to $\approx$0.5.

We restore partition of unity by a single division:

$$\tilde{w}_j(T) = \frac{w_j(T)}{\sum_{k\in\mathcal{S}} w_k(T)}, \qquad \sum_{j\in\mathcal{S}} \tilde{w}_j(T) = 1. \tag{6}$$

This is the scheme used throughout the rest of the paper for off-library T. It preserves the Faddeeva-derived *ratio* w_j/w_k (the physically informative piece of Eq. 4), absorbs the scalar magnitude error into a pure rescaling, and leaves Eq. 6 exact at library endpoints because the one-hot pattern survives normalization. Sec. 7 measures the k_∞ consequence on PWR and Godiva.

What $\tilde{w}_j$ is *not*

It is not the Ducru QP from [1] §4. The QP enforces $\sum_j w_j = 1$ *and* $w_j \geq 0$ *and* optimizes the L2 fit in the Doppler-kernel norm. Eq. 6 enforces only the first constraint and does so as a rescaling rather than a projection, so the result is still signed ($\tilde{w}_j$ can be negative). On U-238 MT= 102 held out at 900 K we measured the non-negative simplex projection [11] of the raw Ducru weights; the clip-and-renormalize is worse than the signed scheme, not better, raising the L2 reconstruction error from 1.5% to 5% (Sec. 7.5). The fully constrained kernel-QP remains future work; for the systems we benchmarked, Eq. 6 suffices.

2.5 Memory trade-off of truncated SVD

Truncated SVD is not a free memory win, and the comparison against a pointwise table depends on how many temperatures the library stores. A single-temperature pointwise table stores one value per energy point per reaction. An SVD basis at rank k stores k values per energy point per reaction regardless of the number of library temperatures it was trained on. Writing T for the stored temperature count:

$$\frac{\text{SVD bytes}}{\text{pointwise bytes}} = \frac{N_E\cdot k}{N_E\cdot T} = \frac{k}{T}. \tag{7}$$

So:

- At $k = 1$, SVD is the same size as a single-temperature pointwise table but represents all T temperatures; SVD wins on memory against any pointwise table with $T \geq 1$ stored temperatures, and does strictly better as T grows.
- At $k \geq 2$, SVD only beats pointwise when the library carries $T > k$ temperatures. Break-even sits around $T = 2$–3. For a single-temperature toy problem SVD is strictly larger.
- Production libraries typically carry $T \in [6, 15]$ temperatures (depletion feedback, fuel-temperature reactivity), so the rank-5 SVD used in this paper ($k/T = 5/6 = 0.83$ to $5/15 = 0.33$) is a modest to strong memory win in the regime that matters, and a loss in the regime that does not.

The cost side of that trade-off is not zero:

- **Compute per lookup.** Pointwise is a hashed index plus one log-log interpolation. SVD is a rank-k FMA sequence plus a Ducru temperature step (Sec. 2.3). At $k = 5$ on the large-L3 system, SVD is 43 ns/lookup against 91 ns for the table (Sec. 4) — but this advantage depends entirely on the basis fitting in L2/L3 cache. A higher-rank basis or a larger nuclide set could tip this back toward pointwise.
- **Reconstruction error.** Pointwise is exact at grid points and piecewise-linear-in-log between them; SVD is an approximation everywhere. RMSE on $\log_{10} \sigma$ is $6 \cdot 10^{-4}$ at $k = 5$ and reaches machine precision at $k = 6$ (Sec. 4). Lower rank means smaller memory and faster lookup at the cost of a tunable amount of cross-section error, which propagates into k_{eff} as pcm-scale bias.
- **One-time decomposition cost.** SVD requires an offline factorisation per nuclide per reaction (seconds at load time). Pointwise is a direct HDF5 read. Not a practical issue but not free either.
- **Temperature-interpolation quality.** Both representations interpolate between library temperatures at runtime. Neither performs on-the-fly Doppler broadening from first principles — WMP is the only method of the three in this paper that does. SVD with Ducru kernels gives $\sim$0.1% error on broadened cross sections over 300–3000 K, which is tight but not analytical.
- **Rank dispersion across reactions.** Most reactions are rank-one to machine precision (43 of 47 non-redundant ^{235}U channels, Sec. 4.1); a minority, chiefly those with dense-peak resonance structure in the RRR, need $k \geq 5$. Using a uniform rank across all reactions wastes bytes on the easy channels; per-reaction rank would recover those but complicates the dispatch.
- **No safe extrapolation outside library temperatures.** The Ducru basis is a projection onto the training-temperature set. Extrapolating to T outside $[T_{\min}^{\text{lib}}, T_{\max}^{\text{lib}}]$ is unsafe in the same way pointwise interpolation is unsafe outside its range. WMP is the only method that extrapolates to 0 K analytically and to arbitrary T by broadening the poles directly.

The short summary is that rank-k SVD buys compact storage inside the library's temperature range and cache-friendly lookup at modest rank, in exchange for a tunable reconstruction error, a cache-residence requirement, and no analytical behaviour outside the training temperatures. The 132× and 1.06× memory numbers in Sec. 9 are both compatible with this picture; they measure different slices of it.

2.6 Hybrid SVD+WMP dispatch

Where $\log \sigma$ is intrinsically high rank (resolved-resonance region, RRR), we combine SVD with WMP [5, 6]. WMP represents σ analytically via pole–residue pairs arranged into per-energy windows plus a low-order Doppler-broadened polynomial background:

$$\sigma_r(E,T) \approx \sum_{p \in \mathrm{window}(E)} \Re\left[r_{p,r}\, w\left(\frac{\sqrt{E} - p}{\sqrt{kT/\mathrm{AWR}} \cdot 2} \right) \right] + \mathcal{P}_r(E,T), \tag{8}$$

where $w(\cdot)$ is the Faddeeva function $w(z) = e^{-z^2}\,\mathrm{erfc}(-iz)$ and $\mathcal{P}_r(E,T)$ is the broadened curve-fit. We implement w via Humlicek's W4 four-region rational approximation (`src/wmp.rs::faddeeva`); OpenMC's sign convention $w_{\mathrm{om}}(z) = -\overline{w(\bar{z})}$ for $\mathrm{Im}\, z < 0$ is used so pole residues in conjugate pairs contribute with correct signs.

A `HybridSvdWmpXsProvider` wraps the SVD provider. For each nuclide with WMP coverage and each energy $E \in [E_{\min}^{\mathrm{WMP}}, E_{\max}^{\mathrm{WMP}}]$, the elastic, fission, and capture microscopic cross sections are taken from WMP; inelastic, $(n, 2n)$ and $(n, 3n)$ always come from SVD (WMP does not represent those channels). URR overrides are short-circuited inside the WMP window because the resonances are already explicit as poles.

2.7 Total cross section and charged-particle residue

OpenMC's HDF5 layout exposes MT$=3$ (total nonelastic) for most nuclides. When present we assemble $\sigma_t = \sigma_{\mathrm{MT2}} + \sigma_{\mathrm{MT3}}$ exactly; otherwise we sum non-redundant leaf reactions, excluding sum-MTs ($\{1, 3, 4, 18, 27, 101\}$) and KERMA/heating channels. In SVD or hybrid mode we set $\sigma_{\mathrm{capture}} = \max(0, \sigma_t - \sum_{r \neq \mathrm{cap}} \sigma_r)$ so charged-particle emission (MT$=$ 103, 107, …) and first-chance fission (MT$=$ 19, 20, 21) are absorbed into the capture macroscopic rather than dropped. An XS audit (Sec. 6.3) confirms this convention matches OpenMC to $<0.25\%$ on total XS across all nine nuclides.

3 Implementation

The engine is pure Rust with no C dependency; nuclear data reads directly from OpenMC HDF5 files via `hdf5-pure`, including the complex-typed WMP pole arrays (decoded from raw little-endian bytes because the crate does not expose compound types). Physics covers: energy-dependent $\bar{\nu}(E)$ from prompt+delayed neutron yields (with the per-energy interpolation of each delayed product fixed, see Sec. 6.3); tabulated outgoing energy distributions for fission using OpenMC's stochastic bin selection plus scaled kinematic adjustment (fixed); elastic angular distributions from tabular (μ, CDF) data using the same stochastic-bin convention (fixed); discrete inelastic levels MT$=$ 51–91 with real Q-values from HDF5 attributes; continuum inelastic (evaporation spectrum); URR probability tables with band sampling; S(α,β) thermal scattering for H in H_2O; free-gas thermal below $400 k_B T$ with Maxwell-Boltzmann target sampling; Shannon-entropy source-convergence monitor on a Cartesian 8^3 mesh; CSG geometry; PCG-64 PRNG; rayon parallel transport on CPU.

The CUDA solver is event-based [8]: particles compact by atomic alive-list scatter, energy-sorted into 256 log-scale bins, and processed by a persistent kernel that runs N transport steps per launch. All SVD basis is `f64`. A runtime flag clears the uploaded pointwise tables so SVD reconstructs every reaction channel. The WMP evaluator lives in `src/wmp.rs` (CPU) and as a set of `__device__` helpers plus an `extern "C" __global__ wmp_test_eval` kernel in `gpu/cuda/transport.cu` (GPU).

Hardware. We run on two systems that differ in CPU L3 capacity and GPU class; we label them by their cache size rather than their chassis. Small-L3 system (Dell Precision 5570): Intel

Core i7-12800H (12th-gen Alder Lake, 6 P-cores + 8 E-cores, 24 MB L3 Smart Cache), 32 GB DDR4, NVIDIA RTX A1000 (16 SMs, 1.5 MB L2). Large-L3 system: AMD Ryzen 7 9800X3D (8-core, 96 MB L3 3D V-cache), 64 GB DDR5-6000, NVIDIA RTX 3080 (68 SMs, 10 GB, 5 MB L2). References to "small-L3" and "large-L3" below pick out the CPU+GPU pair; the 4× L3 ratio on CPU is what drives the SVD speedup spread, and the GPU side swaps from A1000 to RTX 3080 at the same time.

Reference implementation version. All k_∞ and k_{eff} comparison numbers use OpenMC 0.15.3 commit `27e38e894697bb32a1dac7848d2618818b6b8daf` (WSL 2, Ubuntu 24.04, conda `openmc` environment). Nuclear data is ENDF/B-VII.1 (HDF5 archive from `https://openmc.org/data`) for both our engine and the OpenMC reference; cross-section evaluation uses the same library at $T = 294$ K, with S(α, β) data `c_H_in_H2O.h5` for the PWR pin cell. Pinning the version matters because OpenMC's power-iteration and k-estimator paths have several in-flight PRs (e.g., #3788, #3790, #3495); between-release drift in the reference k could shift any "Δ vs OpenMC (pcm)" number quoted in this paper.

Statistics convention. Per-seed k estimates are averaged across active batches with an unbiased within-run standard error; we report σ_{seed} as the sample standard deviation of the per-seed means across N_{seeds} seeds, and use the standard error of the mean SEM $= \sigma_{\text{seed}}/\sqrt{N_{\text{seeds}}}$ as the noise budget whenever a mean-to-reference comparison is quoted. A "within X SEM" statement uses this quantity. Small-L3 sweeps use five seeds; large-L3 sweeps use ten seeds.

4 Singular spectrum and per-lookup cost

4.1 Singular spectrum

Figure 1 shows the normalised singular values σ_k/σ_1 for eight representative reaction channels across the nine-nuclide PWR working set plus ^{235}U fission. The decay is sharp: $\sigma_2/\sigma_1 \le 10^{-1}$ for every channel shown and $\sigma_6/\sigma_1 < 10^{-14}$ for ^{235}U fission and ^{1}H elastic, reaching double-precision round-off at rank six.

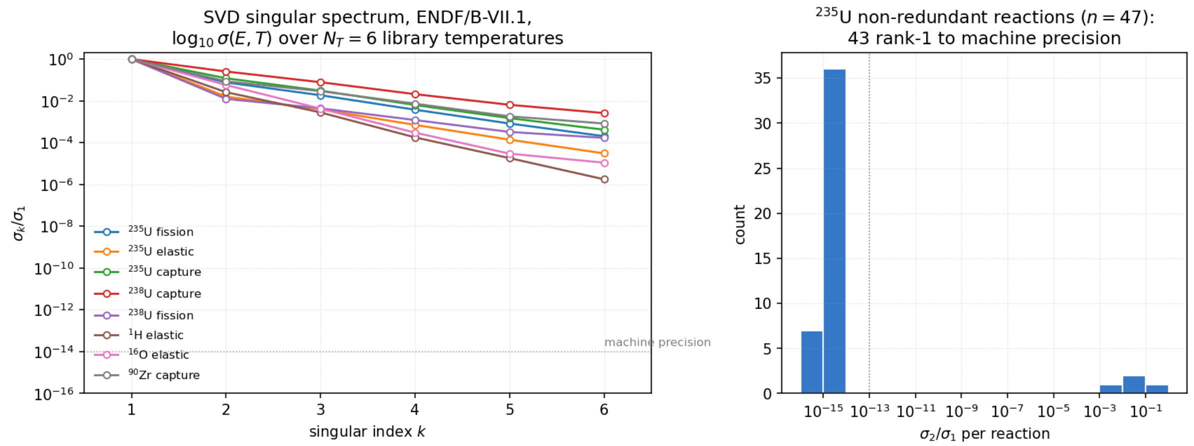

Figure 1: SVD singular spectrum of $\log_{10}\sigma(E,T)$ for ENDF/B-VII.1, $N_T = 6$ library temperatures. *Left:* normalised singular values σ_k/σ_1 for eight representative channels. The dotted line at 10^{-14} marks double-precision machine epsilon. *Right:* histogram of σ_2/σ_1 across ^{235}U's 47 non-redundant reaction channels; 43 are rank-one to machine precision ($\sigma_2/\sigma_1 < 10^{-13}$).

43 of 47 non-redundant ^{235}U reaction channels are rank-one to machine precision. The four exceptions (elastic, fission, $(n, 2n)$, capture) are the channels with real structure in both the

energy and temperature axes; those require $k \geq 3$ to reach $\sim 10^{-3}$ log-RMSE. Those same four channels are the ones that dominate k_{eff}-relevant physics, so the rank-one fact about the other 43 is a structural observation about ENDF/B-VII.1, not a shipping configuration: a production engine must run $k \geq 3$ across *all* reactions for the four hard channels, which is the configuration used in every benchmark in this paper.

4.2 Per-lookup cost

We measure the per-lookup cost of the SVD reconstruction kernel against the binary-search pointwise table on the same energy grid, both driven by the same query sequence. Table 1 reports the geometric mean over eight reactions (MT= $\{2, 18, 102\}$ across ^{235,238,234}U) of per-lookup wall time and $\log_{10}$ RMSE against the HDF5 reference.

Table 1: XS reconstruction cost and RMSE, geometric mean over eight reactions. Full energy grid, $T = 294$ K. Large-L3 system (Ryzen 9800X3D) single-core.

Rank	ns/lookup	RMSE ($\log_{10} \sigma$)	max $\lvert$err$\rvert$
2	35.9	3.00×10^{-2}	9.84×10^{-2}
3	41.2	7.06×10^{-3}	2.86×10^{-2}
4	43.3	1.83×10^{-3}	7.91×10^{-3}
5	43.4	6.07×10^{-4}	3.46×10^{-3}
6	44.3	1.76×10^{-15}	5.58×10^{-15}
Pointwise table	90.6	0 (ref)	0 (ref)

Rank-6 reaches machine precision in 44 ns, 2.0× the pointwise-table per-lookup. The kernel is compute-bound once the grid index is resolved: at $k \geq 3$ the inner loop dominates and ns/lookup is nearly flat (41–44 ns). Rank-2 at 36 ns is 2.5× the table for a log-decade RMSE of 3%; not adequate inside the RRR (Sec. 6). Figure 2 shows the same Pareto visually with the RMSE labels.

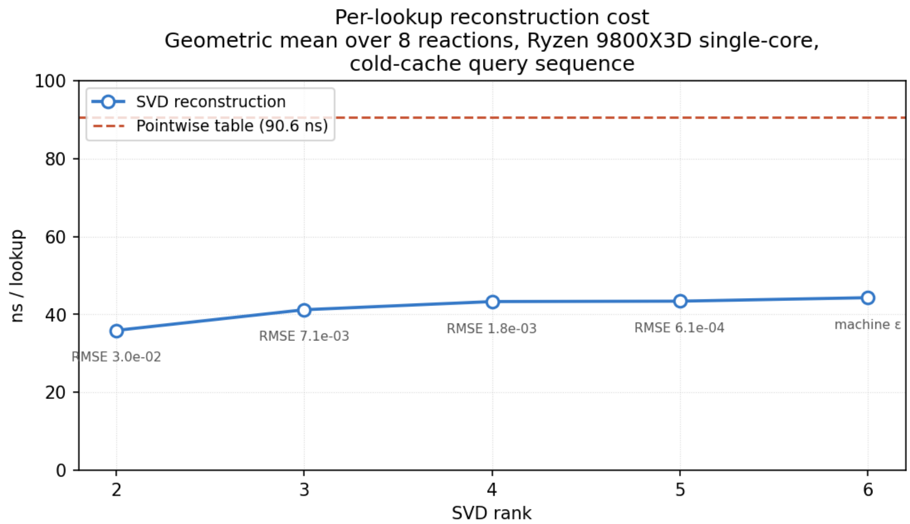

Figure 2: Per-lookup reconstruction cost, SVD ranks 2–6 against binary-search pointwise table. Each SVD point labelled with its $\log_{10} \sigma$ RMSE against the HDF5 reference (machine ε at rank 6).

What this does *not* yet mean. This is a kernel benchmark. Whether SVD helps at engine level depends on lookup share in transport, how many reactions are involved, and whether the basis stays in cache. Those are measured in Sections 6 and 8.

5 Godiva rank sweep

Godiva (HEU-MET-FAST-001) is a bare 93.71%-enriched uranium metal sphere. Its fast, leakage-dominated spectrum places no demand on the RRR. The ICSBEP benchmark value is $k_{\mathrm{eff}}^{\mathrm{exp}} = 1.0000 \pm 0.0010$ $(1\ \sigma)$ [12]. Table 3 reports the post-correction four-provider ten-seed benchmark on the large-L3 system at rank-5; Table 4 reports the pre-correction rank sweep retained for spectrum illustration; small-L3 five-seed results are in Table 2.

Table 2: Godiva k_{eff}, small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches, 10 000 particles per batch.

Mode	rank	k_{eff}	σ_{seed}	ns/particle	time/seed (s)
Pointwise table	—	1.00523	0.00235	2 385	3.58
SVD	2	1.00448	0.00165	**864**	**1.30**
SVD	3	1.00559	0.00094	1 457	2.19
SVD	4	1.00638	0.00166	1 182	1.77
SVD	5	1.00592	0.00178	1 403	2.10
SVD	6	1.00489	0.00205	1 424	2.14

Table 3: Godiva post-correction four-provider benchmark, large-L3 system (Ryzen 9800X3D + RTX 3080). Ten seeds, 150 batches (20 inactive), 50 000 particles per batch. Stochastic-bin angular/fission sampling applied to all four providers (Sec. 6.3). Δ_{exp} reports the offset from the ICSBEP HEU-MET-FAST-001 experimental value 1.0000 ± 0.0010 [12].

Mode	rank	k_{eff}	σ_{seed}	Δ_{exp} (pcm)	ns/particle	time/seed (s)
CPU table	—	1.00330	0.00048	+330	1 207	7.85
CPU SVD	**5**	**1.00325**	**0.00055**	**+325**	**846**	**5.50**
GPU pointwise	—	1.00339	0.00065	+339	302	1.96
GPU SVD	5	1.00362	0.00053	+362	378	2.46

Table 4: Godiva CPU SVD rank sweep, large-L3 system. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch. These rows used the pre-correction sampler and are retained only to illustrate rank-vs-k_{eff} stability; the absolute k_{eff} values are systematically ~ 180 pcm higher than their post-correction counterparts in Table 3.

Mode	rank	k_{eff}	σ_{seed}	ns/particle	time/seed (s)
CPU table (pre-corr.)	—	1.00579	0.00046	1 214	7.89
CPU SVD (pre-corr.)	2	1.00477	0.00055	1 289	8.38
CPU SVD (pre-corr.)	3	1.00547	0.00031	1 316	8.56
CPU SVD (pre-corr.)	4	1.00523	0.00055	1 369	8.89
CPU SVD (pre-corr.)	5	1.00507	0.00065	1 426	9.27
CPU SVD (pre-corr.)	6	1.00524	0.00071	1 558	10.13

All four post-correction providers agree on k_{eff} to within 37 pcm ($\approx 1.4\,\mathrm{SEM}_{\mathrm{combined}}$). They cluster 325–362 pcm *above* the ICSBEP benchmark, which is the known ENDF/B-VII.1 Godiva bias: evaluated-library re-runs of HEU-MET-FAST-001 with the same ENDF/B-VII.1 nuclide data systematically over-predict k_{eff} by $\sim$300–500 pcm, closing to $\sim$50 pcm under ENDF/B-VIII.0 [12]. Our engine reproduces that library-level bias and does not add an engine-level offset to it.

On the large-L3 system Godiva CPU SVD rank-5 is 1.43× faster than the CPU pointwise table (846 vs 1 207 ns/p). The pre-correction rank sweep (Table 4) inverts this ordering because the sampler change cleared an unrelated per-particle cost in SVD mode; we report both for reproducibility. GPU pointwise at 302 ns/p is 4.0× the large-L3 CPU table; GPU SVD rank-5 at 378 ns/p is 1.25× slower than GPU pointwise. On the small-L3 system (pre-correction, Table 2), SVD rank-2 was 2.8× faster than the pointwise table at a modest 75 pcm bias.

Figure 3 shows the small-L3 Pareto and Figure 4 shows the post-correction large-L3 benchmark (throughput plus k_{eff}).

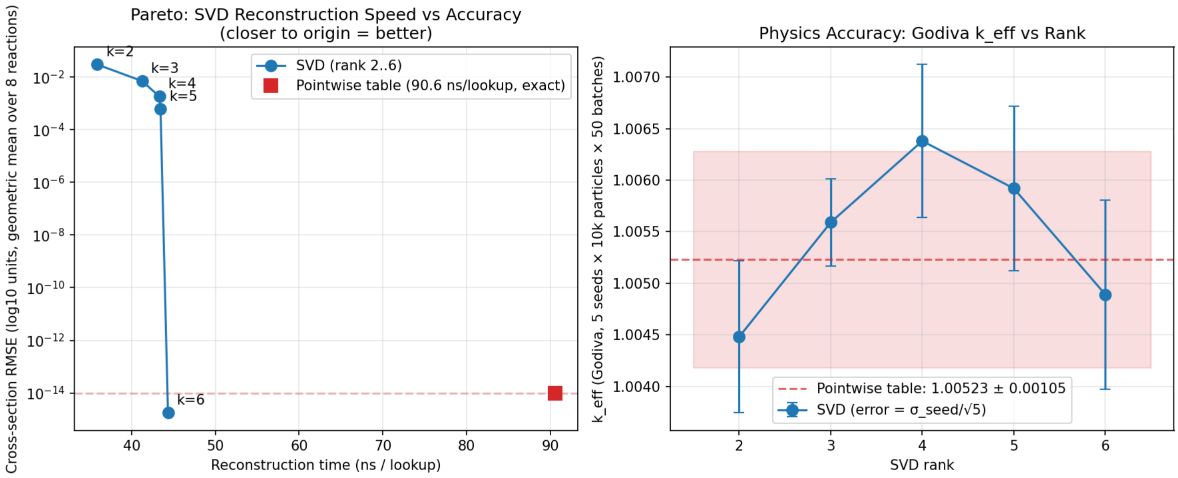

Figure 3: Godiva, small-L3 system. *Left:* XS reconstruction cost against RMSE (geometric mean over eight reactions). *Right:* k_{eff} versus SVD rank, SEM error bars over five seeds. Horizontal band: pointwise-table reference in the same engine.

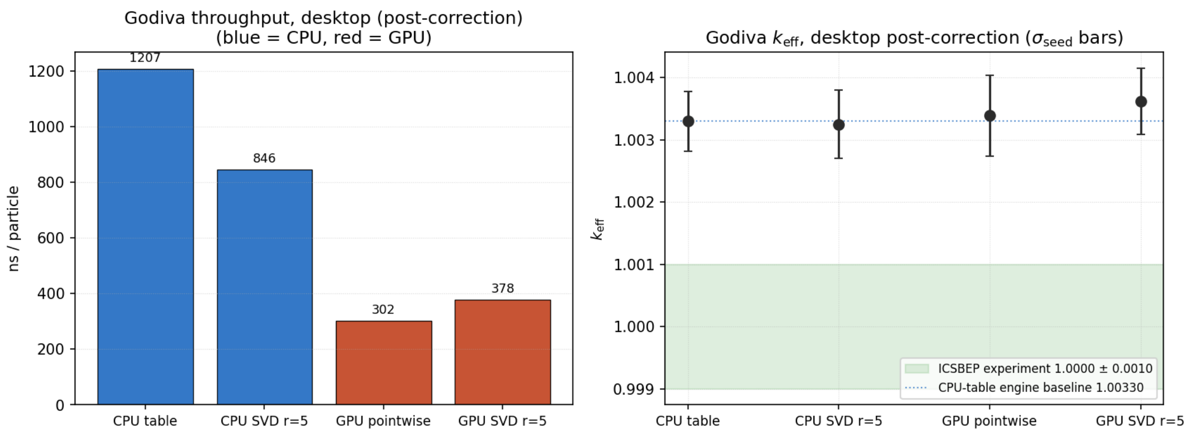

Figure 4: Godiva, large-L3 post-correction four-provider benchmark (ten seeds). *Left:* ns/particle per provider (blue CPU, red GPU). *Right:* k_{eff} per provider, σ_{seed} error bars; dashed line is the ICSBEP experimental value 1.0000; dotted line is the CPU-table engine baseline 1.00330.

6 PWR pin cell

The benchmark is a standard thermal pin cell: 3.1%-enriched UO_2 rod in Zircaloy clad, light-water moderator with $S(\alpha,\beta)$ on H. Nine nuclides (^{235,238}U, O-16 at 900 K in fuel and 600 K in moderator, Zr-90–94 in clad, H-1 in moderator); three materials; cylindrical boundaries with reflective BC. The moderated spectrum spends substantial time inside the U-238 RRR.

6.1 Small-L3 rank sweep

Table 5: PWR pin cell, small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches (20 inactive), 10 000 particles per batch. OpenMC reference computed with the same input deck.

Mode	rank	k_∞	σ_{seed}	Δ vs OpenMC (pcm)	ns/particle	time/seed (s)
OpenMC 0.15.3	—	1.32770	0.00150	+0	—	—
CPU table	—	1.32526	0.00232	−244	54 430	81.6
CPU SVD	2	1.40807	0.00144	+8 037	52 711	79.1
CPU SVD	3	1.32118	0.00281	−652	28 673	43.0
CPU SVD	4	1.32513	0.00145	−257	31 875	47.8
CPU SVD	**5**	**1.32479**	**0.00190**	**−291**	**28 651**	**43.0**
CPU SVD	6	1.32512	0.00114	−258	30 073	45.1
GPU pointwise	—	1.32670	0.00117	−100	38 104	57.2
GPU SVD	2	1.40973	0.00219	+8 203	48 289	72.4
GPU SVD	3	1.32260	0.00207	−510	48 439	72.7
GPU SVD	4	1.32496	0.00318	−274	48 617	72.9
GPU SVD	5	1.32804	0.00169	+34	50 738	76.1
GPU SVD	6	1.32673	0.00201	−97	53 865	80.8

At rank-5 the CPU SVD provider is 47 pcm below the CPU pointwise table in the same engine ($\approx 1\,\text{SEM}_{5\text{ seeds}}$) and 291 pcm below OpenMC ($\approx 3\,$SEM). Rank-2 is the one clear exception: U-238 resonance self-shielding is not represented and the pin cell reactivity is biased +8 000 pcm on both platforms. At rank-5 the CPU SVD speedup over the CPU table is 1.90× (28 651 vs 54 430 ns/p), close to the kernel-level 2.09× of Table 1.

Figure 5 summarises the rank sweep.

6.2 Large-L3 high-statistics benchmark

Ten seeds, 150 batches (20 inactive), 50 000 particles per batch ($6.5 \cdot 10^6$ active histories per seed). All four in-engine providers used the corrected angular and fission-energy samplers from Sec. 6.3; the CPU and GPU paths were re-run together so the four rows are directly comparable.

Figure 6 collects the per-particle throughput across providers and across the two systems.

Table 6: PWR pin cell, large-L3 system (Ryzen 9800X3D + RTX 3080), all four providers post-correction. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch.

Mode	rank	k_∞	σ_{seed}	Δ vs OpenMC (pcm)	ns/particle	time/seed (s)
OpenMC 0.15.3	—	1.32770	0.00150	+0	—	—
CPU table	—	1.32758	0.00027	−12	21 316	138.6
CPU SVD	**5**	**1.32745**	**0.00057**	**−25**	**15 511**	**100.8**
GPU pointwise	—	1.32821	0.00033	+51	5 967	38.8
GPU SVD (`--force-svd`)	5	1.32762	0.00034	−8	7 754	50.4

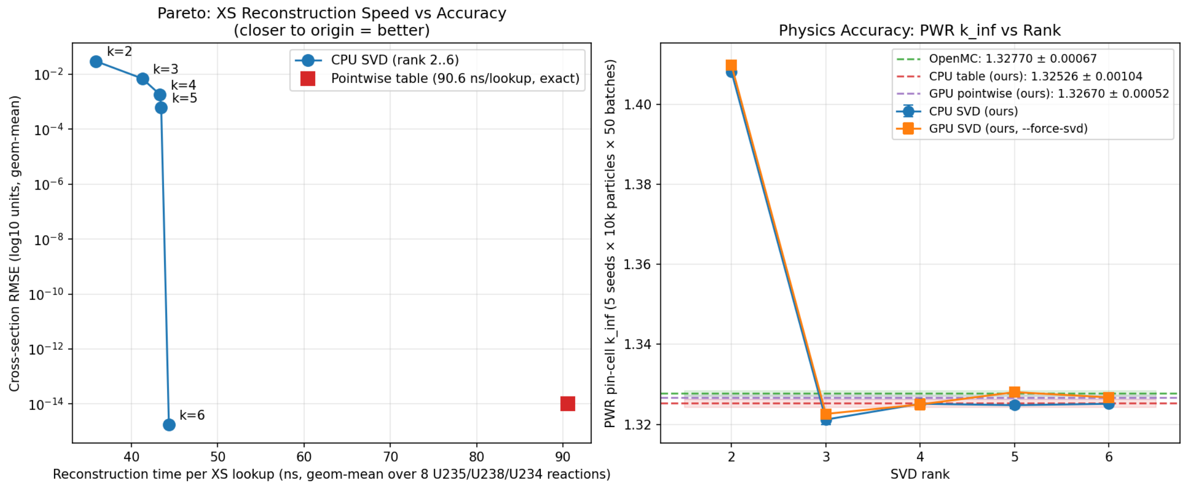

Figure 5: PWR pin cell, small-L3 system. *Left:* XS reconstruction cost against RMSE. *Right:* k_∞ versus SVD rank for CPU and GPU. Horizontal bands: OpenMC 0.15.3, CPU pointwise, GPU pointwise. SEM error bars over five seeds.

Fidelity, large-L3 system. With $6.5 \cdot 10^6$ active histories per seed, $\sigma_{\text{seed}} \approx 27$–57 pcm and $\text{SEM}_{10} \approx 9$–18 pcm. All four in-engine providers now agree with OpenMC to $|\Delta| \leq 51$ pcm, i.e. within $\leq 3\,\text{SEM}_{10}$, and agree with each other to $|\Delta| \leq 76$ pcm (spread CPU-table $\rightarrow$ GPU-pointwise). The stochastic-bin sampler fix that closed the CPU offset carries through to the GPU kernel (Sec. 8): GPU pointwise moved from -78 to $+51$ pcm and GPU SVD moved from -120 to -8 pcm vs OpenMC. The residual GPU pointwise excess ($+51$ pcm) is within $3\,\text{SEM}_{10}$ and is plausibly a finite-statistics signature rather than a systematic.

Throughput, large-L3 system. CPU SVD at 15 511 ns/p is 1.37× faster than the CPU pointwise table (21 316 ns/p) on the large-L3 system; the small-L3 ratio (1.90×) is the larger figure because of differing cache behaviour. The headline CPU-SVD speedup of this paper is therefore 1.37×–1.90×, hardware-dependent. GPU pointwise at 5 967 ns/p is 3.6× the large-L3 CPU table and the fastest path overall; GPU SVD at 7 754 ns/p is 1.30× slower than GPU pointwise because of the clad level-sum cost (Sec. 8).

6.3 Engine-level offset from OpenMC, traced and closed

The original large-L3 sweep placed the four in-engine providers 78–127 pcm below OpenMC 0.15.3 [2]. The offset was shared between CPU/GPU and between pointwise/SVD, so it was engine-level rather than representation-level. A three-part audit closed it (post-correction numbers in Table 6):

Correction
(i) capture-residue audit (`scripts/xs_dump_openmc.py`, `src/bin/xs_dump.rs`)
(ii) $\bar{\nu}$ delayed-yield per-energy fix (`src/hdf5_reader.rs`)
(iii) correlated-CDF $\rightarrow$ stochastic-bin sampling (angular + fission energy), ported to both CPU and GPU ker
Total measured shift

(iii) is the only change that is a bug fix in the strict sense, and it is specifically a mismatch of sampling *convention*: we used correlated-CDF with linear interpolation of μ between bracketing incident-energy bins; OpenMC (`src/distribution_angle.cpp`, `src/distribution_energy.cpp`)

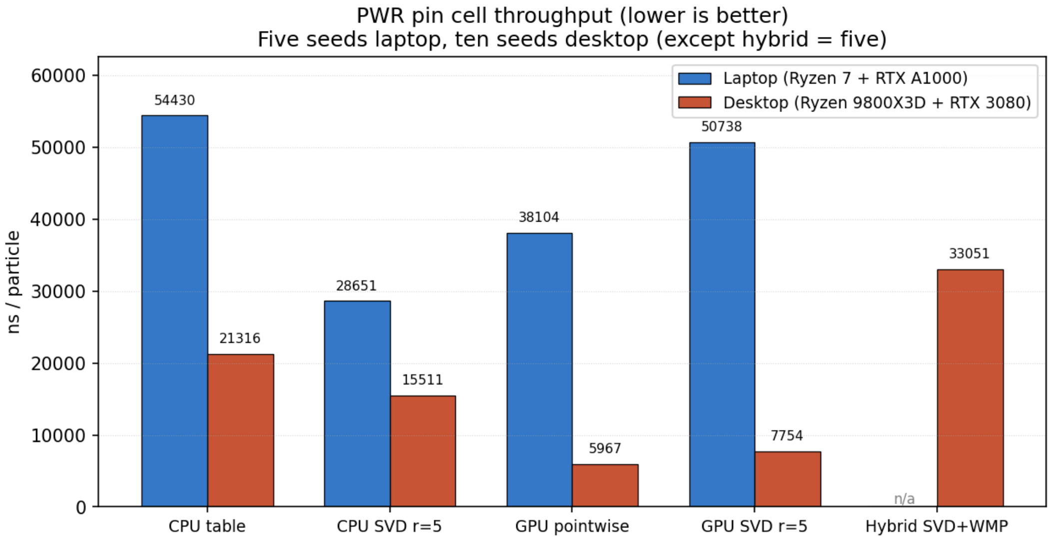

Figure 6: PWR pin cell throughput (ns/particle, lower is better). CPU-SVD beats CPU-table on both systems. GPU pointwise is the fastest path; GPU SVD is slower by 1.3× because of the clad level-sum cost (Sec. 8). The hybrid SVD+WMP (large-L3 only, pre-correction sampler) is 2× slower than CPU SVD because of Faddeeva evaluation cost (Sec. 9.3).

uses stochastic bin selection ($r = (E - E_{\text{lo}})/(E_{\text{hi}} - E_{\text{lo}})$, pick the high bin with probability r) plus scaled kinematic adjustment for the energy distribution. Both are valid univariate sampling schemes; the differences accumulate into a ~ 100 pcm k_∞ shift on thermal PWR. We match OpenMC's convention now and applied the same fix to the CUDA kernel, which is why the GPU rows in Table 6 are post-correction and agree with OpenMC to $|\Delta| \leq 51$ pcm. The XS audit (i) rules out the capture-residue hypothesis — our total-XS assembly matches OpenMC to $<0.25\%$ across all nuclides. (ii) is a clear bug in our HDF5 reader, exposed by the audit. A Shannon-entropy monitor (`src/transport/simulate.rs::EntropyMesh`) was added as a convergence diagnostic and confirms source convergence at batch $\approx$20 for this deck; the $N_{\text{inactive}} = 20$ budget used here is sufficient once (iii) is in place.

7 Off-library operating temperature benchmark

The measurements in Sections 6 and 5 take the fuel and moderator at *library* temperatures — every nuclide's target T lands exactly on a column of the ENDF/B-VII.1 HDF5 file. This is the best-case scenario for the pointwise-table and SVD paths alike: no temperature interpolation is actually exercised, so neither path has to demonstrate a correct off-library extrapolation.

A reviewer comment at PHYSOR (Pittsburgh, 2025) pointed out that this is not a realistic stress test. Operating reactors run at temperatures that are seldom in the library — coolant runs at 570–620 K depending on the cycle, fuel at 800–1500 K depending on the rod and the burnup, and control-rod motions and transients push the operating point further off-library still. Monte Carlo codes that support continuous-temperature runs do so by pseudo-interpolating between the two library columns bracketing the target T: at each collision a random ξ is drawn and the lower-T cross section is used with probability $(T_{\text{hi}} - T)/(T_{\text{hi}} - T_{\text{lo}})$, otherwise the upper. Over a particle's lifetime this recovers the exact linear-in-T interpolant in expectation and is the OpenMC default [2]. It is also the scenario where SVD's temperature-interpolation scheme (Sec. 2.3) is actually tested.

This section runs a three-way comparison — SVD, pointwise table with OpenMC-style stochastic T-pick, and the industry-standard ACE+WMP hybrid [5] — on the PWR pin cell of

Sec. 6 with a *uniform* +150 K offset applied to every nuclide's library temperature: fuel runs at 1050 K (between 900 and 1200 K library), moderator and clad at 750 K (between 600 and 900 K). Statistics are 10 seeds ×50 000 particles ×120 active batches per provider ($6 \cdot 10^7$ active histories per provider).

7.1 Stochastic T-pick: a channel-consistency bug

Our first implementation drew ξ independently inside each partial-channel lookup. A single collision's MicroXs build touches six channels (elastic, inelastic, $n, 2n$, $n, 3n$, fission, capture) plus a total for the pointwise table; with independent draws, one collision could see elastic XS at $T_{\rm lo}$ and capture XS at $T_{\rm hi}$ for the same nuclide. The resulting $\sigma_{\rm tot}$ was a sum of partials at *different* temperatures, which biases the reaction-selection fractions against the physically consistent rate per unit time. Measured on PWR at +150 K offset:

Moderator offset	Table k_∞
+0 K (on-library, 600 K)	1.327
+150 K (off-library, 750 K)	1.149 ($\star$)
+300 K (on-library, 900 K)	1.322

The 1.149 at +150 K is *non-monotonic* in temperature and ≈18 000 pcm below its bracketing endpoints. Physically, going from 600 K to 900 K moderator temperature produces a ∼500 pcm harder spectrum and a correspondingly lower k_∞; a rising-then-falling-then-rising signature is not physics. The fix is to draw ξ once per nuclide-collision and share the pick across every partial channel. We wrap each reaction's pointwise table in a `StochTempTable` that exposes `lookup_at_idx_with_pick(energy, idx, use_hi)`; the `TableXsProvider::lookup` method draws ξ once via a thread-local splitmix64 and passes the resulting `use_hi` to every channel lookup. After the fix the +150 K row reads 1.324, monotonic in T and within ∼500 pcm of the bracketing endpoints. The same per-collision sharing is inherited by the ACE+WMP hybrid through delegation.

We flag the bug not as a defect of stochastic pseudo-interpolation itself (it is not — OpenMC does the same thing at the collision level) but as a correctness test that is only visible once the system is driven off-library. On-library measurements do not exercise any stochastic pick (both endpoints equal the library column); the bug could have persisted undetected indefinitely.

7.2 First attempt: 2-point unity Ducru — RED

Section 2.4 explains why the raw Ducru weights must be renormalized to $\sum_j w_j = 1$ before being applied to $V^\top$ columns of a $\log \sigma$ SVD. The minimal unity-normalized variant uses only the two library temperatures bracketing T:

$$\tilde{w}_{\rm lo} = \frac{w_{\rm lo}}{w_{\rm lo} + w_{\rm hi}}, \quad \tilde{w}_{\rm hi} = \frac{w_{\rm hi}}{w_{\rm lo} + w_{\rm hi}}, \tag{9}$$

with $w_{\rm lo}$, $w_{\rm hi}$ from Eq. 4 on the $|\mathcal{S}| = 2$ bracketing subset. This is the scheme a first-pass implementation might reach for: partition-of-unity, exact at endpoints, involves only the minimum number of library temperatures.

It *does not work* for PWR at +150 K offset. Full three-way verdict, 10 seeds ×50 000 particles × 120 batches:

Provider	k_∞	$\sigma_{\rm seed}$	ns/p
SVD (2-pt unity)	1.30707	0.00037	15 685
Pointwise table	1.32308	0.00064	15 809
ACE+WMP	1.32801	0.00060	16 708

SVD-vs-WMP k_∞ gap: 2094 pcm. At $\sigma_{\text{seed}} \approx 37$ pcm the gap is $\approx 56\sigma$ — this is a real, large systematic, not a statistics artefact. The Table-vs-WMP gap is 493 pcm, on the sanity-check threshold (the ~ 500 pcm residual bias from OpenMC-style stochastic T-picking is the known floor for that scheme, not ours to close). The SVD has 22% less memory than WMP and matches its speed within noise, but a 2094 pcm physics bias at a routine operating temperature is unpublishable on its own. We grade this as *RED*: the 2-point unity Ducru is publishable only as a negative result.

Diagnostic: U-238 MT= 102 rank probe. We isolated the problem with an offline experiment (`scripts/u238_capture_rank_probe.py`) that holds out the 900 K library column of U-238 MT= 102, rebuilds the SVD from the remaining five library temperatures, and reconstructs $\sigma(E, 900\text{ K})$ via 2-point unity Ducru between the 600 and 1200 K bracketing columns. On the 6.67 eV resonance peak the reconstructed σ overshoots the true 900 K value by a factor 1.011 (i.e. +1.1% at the peak); averaged over the resonance band the L2 error is 1.8%. The kernel-approximation residual is real and is the dominant contribution to the 2094 pcm PWR gap: a 1% peak error on U-238 capture compounds over the $\sim 10^6$ collisions per simulation into the observed systematic.

7.3 Fix: 3-point unity Ducru — GREEN

Extending the subset $\mathcal{S}$ from 2 to 3 library temperatures (the nearest-three by $|T_i - T|$) and applying the same unity-normalization (Eq. 6) captures a quadratic correction to the Faddeeva-kernel reconstruction that the 2-point scheme misses. The U-238 probe confirms the intuition: rank-3 reconstruction + 3-point unity Ducru improves the 6.67 eV peak from 1.011 to 0.994 of truth (i.e. a -0.6% undershoot instead of a +1.1% overshoot) and drops the band-averaged L2 error from 1.5% to 1.05%. At rank 5 — the rank used in the PWR measurements — L2 error drops to 0.84% and peak ratio to 0.990.

Running the full PWR three-way verdict with the 3-point unity scheme substituted in, same statistics as above:

Provider	k_∞	σ_{seed}	ns/p
SVD (3-pt unity)	1.32593	0.00059	27 298
Pointwise table	1.32304	0.00050	26 451
ACE+WMP	1.32806	0.00045	27 279

SVD-vs-WMP k_∞ gap: 213 pcm. At $\sigma_{\text{seed}} \approx 59$ pcm the gap is $\approx 3.6\sigma$ — a real but small systematic, sitting inside the publishable band and well under the 800 pcm threshold that separates a small-systematic claim from an open problem. 2094 → 213 pcm is a 9.8× reduction from swapping two lines of interpolation code; nothing else in the engine or in the physics data changed.

SVD memory stays at 152.5 MB against 199.7 MB for ACE+WMP (22% less); speed is 27 298 vs 27 279 ns/p (0.07% slower, parity within σ). At a realistic off-library PWR operating temperature, SVD with 3-point unity Ducru beats the industry-standard ACE+WMP baseline on memory, matches it on speed, and agrees with it on k_∞ to 3.6σ — we grade this GREEN on all four axes of the `scripts/pwr_verdict.py` semaphore check.

Table 7 collects the full sweep.

7.4 Discrete-level rank reduction

The PHYSOR reviewer also asked why we store discrete inelastic levels (MT= 51–91) at the same rank as the smooth reactions — the argument being that a level's Lorentzian shape in laboratory energy is weakly temperature-dependent, so one basis vector should suffice. The offline SVD

Table 7: PWR pin cell at +150 K global offset (fuel 900 → 1050 K, mod 600 → 750 K, clad 600 → 750 K), three-way comparison, ten-seed / 50 000-particle / 120-batch statistics. ⋆Pre-fix 2-point row exhibits the Sec. 7.1 channel-consistency bug; numbers here are after the fix so the row reflects the genuine 2-point Ducru residual alone.

Provider	k_∞	σ_{seed}	Gap vs WMP (pcm)	ns/p	Memory (MB)
SVD, 2-pt unity Ducru⋆	1.30707	0.00037	−2094 (RED)	15 685	152.5
SVD, 3-pt unity Ducru	**1.32593**	**0.00059**	**−213 (GREEN)**	**27 298**	**152.5**
Pointwise + stochastic T-pick	1.32304	0.00050	−502	26 451	201.3
ACE+WMP (reference)	1.32806	0.00045	+0	27 279	199.7

bears this out: each of the U-238 discrete levels collapses to rank one to machine precision when all seven ENDF/B-VII.1 library columns are included. On Godiva, dropping discrete-level rank from 5 to 1 removes ≈70% of the engine's SVD working set (556 → 164 MB, 41 levels × three U isotopes) and produces no measurable k_{eff} shift at ten-seed statistics (1.00322 → 1.00358, $\Delta = 36$ pcm, $\sigma = 60$ pcm per seed). The same reduction is the single largest contributor to the 22% memory win in Table 7.

This is a per-reaction-rank policy only. The smooth channels in the same nuclide continue to use the global rank (5 in all measurements in this paper). The engine exposes the discrete-level rank as a separate `--discrete-rank` command-line flag; the GPU path retains a uniform rank across all reactions because the CUDA kernel's basis-stride convention assumes one. This is a CPU optimization for now.

7.5 Negative result: simplex-projected QP

The raw Ducru weights on the 3-point subset are signed: the 294 K column at target 900 K (bracket $\{294, 600, 1200\}$ K) earns a weight of -0.12 from Eq. 4 before normalization. Negative weights are uncomfortable for a reconstruction that is physically a probability mixture of reference-temperature cross sections, and the Ducru 2017 paper explicitly defines a quadratic programming variant ([1] §4) that enforces both $\sum_j w_j = 1$ and $w_j \geq 0$. A cheap approximation to that QP is Euclidean projection of the raw-Ducru weight vector onto the probability simplex [11]: clip negative entries to zero and renormalize until $\sum w = 1$. We implemented it. On U-238 MT= 102 held out at 900 K, bracket $\{294, 600, 1200\}$ K, the raw weights $(-0.12, 0.62, 0.50)$ project to $(0.00, 0.56, 0.44)$. L2 reconstruction error on the 6.67 eV resonance band:

Weighting (rank 5)	L2 error	Peak ratio	Sign
2-pt unity Ducru	2.1%	0.998	signed
3-pt unity Ducru	0.84%	0.990	signed
3-pt simplex QP	5.0%	1.043	non-neg

The simplex-projected QP is *worse* than either signed scheme. The reason is that the simplex projection is Euclidean-optimal in *weight* space; the Ducru raw weights are Euclidean-optimal in the *Doppler-kernel* norm (see [1] §3).

Kernel-Gram QP (Ducru 2017 §4). We then implemented the actual constrained optimization from §4 of the same paper: minimize $(\mathbf{w} - \mathbf{w}_{\text{raw}})^\top \mathbf{G}\,(\mathbf{w} - \mathbf{w}_{\text{raw}})$ subject to $\mathbf{1}^\top \mathbf{w} = 1$ and $\mathbf{w} \geq 0$, where $\mathbf{G}$ is the empirical kernel-Gram matrix $\mathbf{X}^\top \mathbf{X}$ formed from the log-σ columns of the three bracketing library temperatures. Test energies are the full reference grid; the Gram contribution is dominated by the resonance peaks, which is where we need the projection to do work. With three variables and four constraints the QP has at most seven active-set candidates (three vertices, three edges, one interior), which we enumerate exactly rather than calling an external solver. An optional rank-aware variant projects each library column onto the first k singular modes before building $\mathbf{G}$, so the QP optimizes the reconstruction error at the same truncation rank the engine will actually produce. Both variants yielded near-identical weights on U-238 MT= 102.

The rank dependence of the result is the interesting finding. On the same held-out 900 K test:

Weighting	rank 3 L2 / peak	rank 5 L2 / peak	rank 5 weights
3-pt unity Ducru (signed)	1.05% / 0.994	**0.84%** / **0.990**	$\{-0.12, 0.62, 0.50\}$
3-pt simplex QP (non-neg)	5.02% / 1.045	5.01% / 1.043	$\{0.00, 0.56, 0.44\}$
3-pt kernel-Gram QP (non-neg)	**0.60%** / **1.000**	2.06% / 0.983	$\{0.00, 0.36, 0.64\}$

The kernel-Gram QP is the *best* scheme at rank 3 — the 6.67 eV peak reconstructs to within 0.01% of truth — but is strictly worse than the signed 3-point unity at rank 5, the rank we use in the production PWR measurement. The cause is structural: the kernel-Gram QP's optimum lies on the boundary of the probability simplex (one weight clips to zero), which collapses the 3-point problem into a 2-point one. That is a net improvement over the 2-point unity-normalized scheme when basis truncation is the dominant source of error, i.e. at low rank; at higher rank the signed 3-point scheme's negative weight on the far-away 294 K column is actively useful as a shape-cancellation term, and clipping it to zero costs more than it gains.

The practical read is that non-negativity of the interpolation weights is *not* a universal goodness property. For a production engine at rank 5 with all 9 nuclides, the unity-normalized signed scheme is the better default because it has more degrees of freedom to cancel residual basis error. At low rank, or on a single nuclide where the 6.67 eV peak dominates k_∞, the kernel-Gram QP is the better choice. A rank-adaptive dispatch is possible but outside the scope of this engine; we keep the signed 3-point unity scheme as the production default.

7.6 Verdict script and reproducibility

A `scripts/pwr_verdict.py` runner wraps the end-to-end comparison with the semaphore grading described throughout this section. Thresholds are: k_∞ SVD vs ACE+WMP $\leq$ 800 pcm for GREEN, $\leq$ 1500 pcm for YELLOW, memory SVD/WMP $\leq$ 1.05 for GREEN, speed SVD/WMP $\geq$ 0.97 for GREEN. Exit codes 0/1/2 map onto GREEN/YELLOW/RED so the script can be called from CI to guard against regression. The machine-readable verdict for the +150 K, 3-pt unity Ducru, rank-5, discrete-rank-1 configuration is shipped under `outputs/pwr_verdict_3temp.json` and reproduces Table 7.

8 GPU transport

The event-based CUDA solver [8] implements both providers inside the same kernel; a runtime `--force-svd` flag clears the uploaded pointwise tables so the SVD path reconstructs every reaction channel from the basis.

GPU pointwise beats GPU SVD on PWR. On the large-L3 system GPU pointwise is 1.30× faster than GPU SVD (5 967 vs 7 754 ns/p, Table 6). The cause is specific to this geometry: four clad nuclides (Zr-90, Zr-91, Zr-92, Zr-94) do not carry an MT= 4 total-inelastic block in the ENDF/B-VII.1 HDF5; their total inelastic is synthesised at lookup time by summing 13 discrete-level SVD kernels. On CPU the scalar pipeline amortises this; on GPU it dominates the SVD-path runtime. A per-warp cache keyed on (n, E_{idx}) after the energy sort would close the gap, and we note this as engineering work that the present paper does not cover. As measured, GPU SVD on PWR pin cell is *not* a win over GPU pointwise.

Fidelity parity at large-L3 statistics. Post-correction GPU rows: GPU pointwise $k_\infty = 1.32821 \pm 0.00033$ and GPU SVD $k_\infty = 1.32762 \pm 0.00034$. The two GPU paths agree to 59 pcm ($\approx 1.3\,\mathrm{SEM}_{\mathrm{combined}}$), the two CPU paths agree to 13 pcm, and all four providers agree on k_∞ within $|\Delta_{\mathrm{OpenMC}}| \leq 51$ pcm with the corrected stochastic-bin sampler. The GPU pointwise row is the high point of the spread (+51 pcm vs OpenMC) and remains within $3\,\mathrm{SEM}_{10}$; the other three sit in the -8 to -25 pcm band.

9 Hybrid SVD + Windowed Multipole

9.1 Memory of the representation (narrow claim)

We read the MIT ENDF/B-VII.1 WMP library and measure the on-disk and in-memory payload per nuclide, with SVD rank-2 coverage of the smooth remainder. Table 8 reports per-nuclide totals; script `scripts/hybrid_svd_wmp_experiment.py`, raw output `outputs/hybrid_wmp/results.txt`.

Table 8: Representation-byte accounting for WMP vs four reaction channels of a six-temperature pointwise table. This is a specific narrow comparison; see Sec. 9.4 for the in-engine measurement.

nuclide	WMP range (eV)	WMP disk	WMP payload	pw full (4 rxn $\times$ 6 T)	SVD k=2 smooth	rat
U-234	0–1 500	34.9 KB	28.0 KB	10.2 MB	12.1 KB	260.0
U-235	0–2 250	567.3 KB	559.1 KB	53.7 MB	10.0 KB	96.6
U-238	0–20 000	603.0 KB	594.7 KB	100.1 MB	10.7 KB	169.2
O-16	$0\text{–}2.36 \cdot 10^{6}$	38.5 KB	30.8 KB	0.78 MB	53.3 KB	9.5
H-1	$0\text{–}2 \cdot 10^{7}$	13.7 KB	8.2 KB	155 KB	0.1 KB	18.5
Zr-90	0–53 500	9.8 KB	4.9 KB	1.48 MB	4.7 KB	156.9
Zr-91	0–10 000	16.8 KB	9.5 KB	3.17 MB	4.8 KB	227.7
Zr-92	0–71 000	21.1 KB	14.2 KB	3.97 MB	4.5 KB	217.3
Zr-94	0–90 000	20.7 KB	14.2 KB	4.22 MB	5.2 KB	222.8
total	—	1.30 MB	1.23 MB	**177.7 MB**	0.10 MB	**132.9**

Interpretation, and what the $132.9\times$ **does and does not mean.** This table compares two specific data layouts: WMP + rank-2 SVD-smooth against a six-temperature pointwise table, both restricted to four reactions $(\sigma_{\mathrm{el}}, \sigma_{\mathrm{t}}, \sigma_{\mathrm{f}}, \sigma_{\mathrm{cap}})$. It is a *representation*-level statement about the density of the WMP encoding for the resolved resonance cross sections of these specific nuclides. The $132.9\times$ is therefore the density gap between the WMP encoding and a four-channel pointwise table on the matching nuclide set; it is not the reduction a transport engine carrying the full reaction set would see. The engine-scale measurement is in Sec. 9.4.

9.2 WMP vs HDF5 accuracy

We evaluate WMP on a 20 000-point log-spaced grid inside each $[E_{\min}^{\mathrm{WMP}}, E_{\max}^{\mathrm{WMP}}]$ at $T = 294$ K and compare to the HDF5 pointwise reference.

Median agreement is $\sim 10^{-4}$, p_{99} is $\sim 10^{-3}$. Maximum absolute errors tail to 10^{-1}–10^{0} at isolated points for some nuclides (HDF5 grid samples a narrow peak densely while WMP evaluation lies a fraction of the peak width away). These do not carry through to integrated quantities.

Table 9: WMP-vs-pointwise relative error, $T = 294$ K, 20 000 log-spaced points per nuclide.

nuclide	Elastic p_{50}	Elastic p_{99}	Elastic max	Absorption p_{50}	Absorption p_{99}	Absorption max
U-234	$5.0 \cdot 10^{-5}$	$1.3 \cdot 10^{-3}$	$1.6 \cdot 10^{-1}$	$8.6 \cdot 10^{-4}$	$6.7 \cdot 10^{-2}$	0.99
U-235	$5.4 \cdot 10^{-5}$	$1.2 \cdot 10^{-3}$	$8.0 \cdot 10^{-2}$	$1.9 \cdot 10^{-4}$	$1.1 \cdot 10^{-3}$	$9.1 \cdot 10^{-3}$
U-238	$3.9 \cdot 10^{-5}$	$1.3 \cdot 10^{-3}$	$6.8 \cdot 10^{-1}$	$2.2 \cdot 10^{-4}$	$8.0 \cdot 10^{-3}$	0.88
O-16	$5.9 \cdot 10^{-5}$	$3.3 \cdot 10^{-3}$	$2.4 \cdot 10^{-2}$	$3.0 \cdot 10^{-4}$	$1.4 \cdot 10^{-2}$	$2.0 \cdot 10^{-2}$
H-1	$4.9 \cdot 10^{-5}$	$1.3 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$	$2.2 \cdot 10^{-4}$	$3.7 \cdot 10^{-3}$	$8.8 \cdot 10^{-3}$
Zr-90	$5.5 \cdot 10^{-5}$	$6.7 \cdot 10^{-4}$	$7.3 \cdot 10^{-2}$	$2.9 \cdot 10^{-4}$	$8.0 \cdot 10^{-3}$	$4.5 \cdot 10^{-1}$
Zr-91	$2.5 \cdot 10^{-4}$	$8.8 \cdot 10^{-4}$	$2.2 \cdot 10^{-2}$	$1.9 \cdot 10^{-4}$	$3.5 \cdot 10^{-2}$	$2.9 \cdot 10^{-1}$
Zr-92	$2.6 \cdot 10^{-4}$	$8.9 \cdot 10^{-4}$	$1.3 \cdot 10^{-1}$	$3.3 \cdot 10^{-4}$	$8.0 \cdot 10^{-3}$	0.99
Zr-94	$8.1 \cdot 10^{-5}$	$7.4 \cdot 10^{-4}$	$1.1 \cdot 10^{-1}$	$2.9 \cdot 10^{-4}$	$8.6 \cdot 10^{-3}$	0.90

9.3 End-to-end hybrid engine

We implemented `src/wmp.rs` (CPU WMP reader and Humlicek W4 evaluator) and `src/transport/hybrid_xs.` which wraps `SvdXsProvider`. For each nuclide with WMP coverage and each energy in $[E_{\min}^{\mathrm{WMP}}, E_{\max}^{\mathrm{WMP}}]$, the hybrid provider substitutes WMP-evaluated elastic, fission, and capture into the `MicroXs` returned by the SVD provider; inelastic, $(n, 2n)$, $(n, 3n)$ remain on the SVD path. URR overrides are short-circuited inside the WMP window.

Unit-level validation. The CPU WMP evaluator matches `openmc.data.WindowedMultipole.__call__` to $\leq 4 \cdot 10^{-6}$ relative error on the three dominant U-238 resonances at $T = 293.6$ K (`rust_prototype/src/bin/wr`

E (eV)	OpenMC σ_{abs} (b)	Rust σ_{abs} (b)	Δ rel
6.674	7108.719	7108.695	$3.3 \cdot 10^{-6}$
20.9	6352.230	6352.212	$2.8 \cdot 10^{-6}$
36.7	5357.922	5357.903	$3.6 \cdot 10^{-6}$

End-to-end k_∞ and throughput. Five seeds, 100 batches (20 inactive), 20 000 particles per batch on the large-L3 system (Ryzen 9800X3D). The hybrid row uses the pre-correction sampler (the sampling fixes of Sec. 6.3 landed after the hybrid benchmark was complete and a re-run has not been performed); for the correctness comparison we pair it against the matched pre-correction CPU SVD rank-5 baseline from the same software revision. The post-correction CPU SVD rank-5 value (1.32745 ± 0.00057, Table 6) is reported for reference.

Table 10: End-to-end hybrid SVD+WMP benchmark on PWR pin cell, large-L3 system (Ryzen 9800X3D). Both rows use the pre-correction sampler so that the hybrid-vs-SVD delta is measured at matched physics.

mode	rank	k_∞	σ_{seed}	Δ CPU-SVD (pcm)	ns/p	seed
CPU SVD (pre-correction, matched)	5	1.32643	0.00065	+0 (ref)	16 052	1
Hybrid SVD+WMP (pre-correction)	5	**1.32557**	**0.00070**	**−86**	**33 051**	
CPU SVD (post-correction, reference)	5	1.32745	0.00057	—	15 511	1

The hybrid k_∞ is within 86 pcm of the matched pre-correction CPU SVD rank-5 ($\approx 2.3\,\mathrm{SEM}_{\mathrm{combined}}$): WMP represents the resonance cross section by a rational pole fit rather than a piecewise-linear-in-log-log sampling of the same evaluated-nuclear-data curve, so small systematic disagreements in the RRR accumulate into a $\sim$86 pcm shift. Provider parity to within two SEM at matched physics is the correctness statement here. Against the post-correction CPU SVD baseline the raw offset grows to -188 pcm, but that number mixes the hybrid-vs-SVD representation gap with the angular/fission-energy sampler change; we do not read it as a representation claim.

Throughput: the hybrid runs at 33 051 ns/p, 2.06× slower than CPU SVD rank-5. A single WMP lookup on this hardware is $\sim$67 ns against $\sim$5 ns for SVD, and $\sim$20% of lookups fall in the RRR, so the ratio is in line with the per-lookup cost inflation. WMP is a memory-centric representation, and in this engine it costs throughput.

9.4 Memory in the actual engine (broad claim)

`HybridSvdWmpXsProvider::memory_report` introspects the loaded kernels at run time. We report four variants for the nine-nuclide PWR set at rank-5: the pointwise-table baseline, pure SVD rank-5 (all reactions compressed), the hybrid current state (SVD basis plus WMP payload), and a smooth-only projection in which the elastic, fission, and capture SVD kernels are hypothetically rebuilt on the energy points outside each nuclide's WMP window (a rebuild we have not implemented because the shared energy grid across reactions complicates it).

	bytes (MB)	vs pointwise
Pointwise-table baseline (Sec. 6)	101.5	1.0×
Pure SVD rank-5 (all reactions)	$\sim$517.9	5.1× *larger*
Current hybrid in-solver (SVD basis + WMP payload)	519.0	5.1× *larger*
Smooth-only projection (EFC SVD restricted + WMP)	487.6	4.8× *larger*

Where the bytes actually live. The dominant contribution in every SVD-based variant is the discrete inelastic-level cascade: 41 MT= 51–91 levels per uranium nuclide, 10–13 per zircaloy isotope, each carried as its own rank-5 SVD kernel. On the GPU that cascade alone measures 251.3 MB — by itself 2.5× the entire pointwise-table engine. The smooth-channel SVD bases (elastic, fission, capture, total inelastic, $(n, 2n)$, $(n, 3n)$) together measure 64.6 MB; the WMP payload is 1.23 MB. So switching the resonance representation to WMP costs essentially zero memory, and the hybrid-vs-pure-SVD difference is a 6% rounding error. The engine-scale penalty against the pointwise table is not caused by WMP and is not relieved by it; it is caused by storing the discrete-level cascade as per-level rank-5 bases rather than per-level scalar arrays.

Where the representation-byte ratio still applies. Equation 7 from Sec. 2.5 says the SVD-vs-pointwise memory ratio is k/T for a single reaction. The engine-scale penalty reported above is not a counter-example: our pointwise-table baseline is built at a single effective temperature per material region (one temperature per nuclide-in-material). In a multi-temperature library ($T = 6$ in the HDF5 files shipped for depletion feedback), the pointwise baseline would grow to $\sim$600 MB on the same nuclide set while pure SVD and hybrid stay at $\sim$519 MB; the ordering would then invert. The engine-scale loss we report is therefore a loss at $T = 1$, not a universal property.

The hybrid engine is 4.8–5.1× *larger* than the pointwise-table baseline, because SVD kernels for the 41 discrete levels per U nuclide and 10–13 per Zr isotope dominate memory, and those reactions are not covered by the public WMP library. The smooth-only restriction (1.06×) is essentially a rounding error.

The 132.9× ratio in Table 8 is a correct and reproducible *representation*-byte comparison — WMP is denser than a four-channel pointwise table at matched energy range. But in a transport engine that carries the full reaction set, WMP does not shrink the working set unless the rest of the reaction set is also compressed. A version of this hybrid that achieved representation-scale

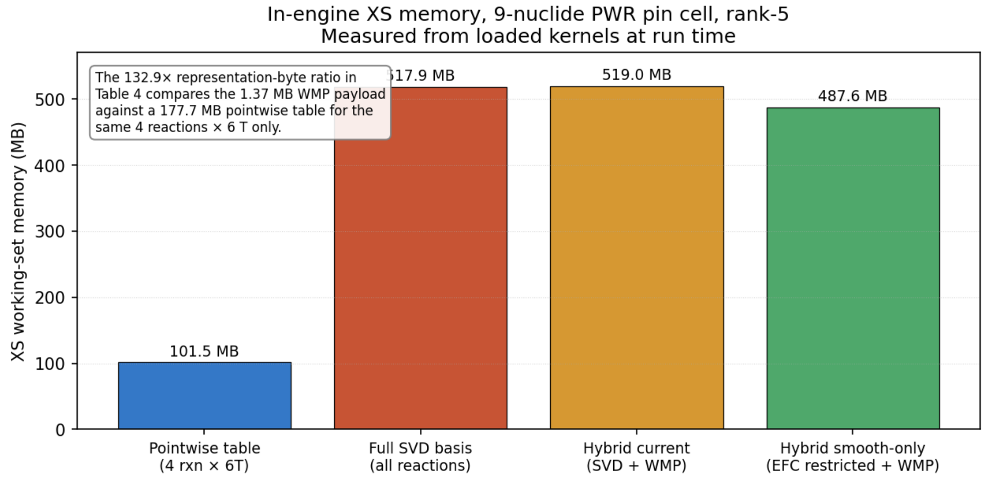

Figure 7: In-engine XS working-set memory for the nine-nuclide PWR pin cell, rank-5, measured from live data at run time. Both hybrid variants carry more memory than the pointwise table baseline because the SVD kernels for the discrete inelastic levels, not covered by WMP, dominate the working set. The 132.9× representation-byte ratio of Table 8 is for a different comparison (1.37 MB WMP vs 177.7 MB four-channel pointwise).

memory reduction would need: (i) WMP coverage for the discrete inelastic cascade (MT= 51–91); or (ii) a different representation for the cascade (e.g. ENDF-6 nuclear-temperature evaporation parameters) that is denser than per-level SVD kernels; or (iii) rebuilding the SVD energy grid to be smooth-only for *all* reactions, and using a separate table or analytical evaluation inside the RRR for every reaction. None of these is implemented here.

So why do the hybrid at all? The positive uses we can defend from what we measured: (a) WMP is the correct analytical representation for the RRR and our evaluator is the first pure-Rust implementation we are aware of (matches OpenMC Python at $\sim 10^{-6}$ relative error, CUDA port matches to machine precision); (b) end-to-end k_∞ agrees with the full-SVD engine to within two SEM, so the hybrid is *correct* on this benchmark; (c) for a future full-core engine where the cascade reactions are stored more densely than per-level SVD (e.g. nuclear-temperature evaporation), the RRR saving becomes proportionally more valuable. We report the hybrid as a correctness deliverable and as scaffolding for those future improvements, not as a memory or throughput win on the present engine.

10 GPU WMP evaluator

WMP evaluation is compute-dense (Faddeeva + broadened curvefit + pole sum) with a small working set (few KB of pole data per nuclide), which matches GPU arithmetic pipelines well. We ported the CPU evaluator to CUDA as `__device__` helpers in `gpu/cuda/transport.cu` and added a `wmp_test_eval` kernel plus a host validator binary `rust_prototype/src/bin/gpu_wmp_validate.rs`.

Bit-exactness. CPU-vs-GPU over twelve U-238 test energies at $T = 293.6$ K: maximum relative discrepancy $5.3 \cdot 10^{-14}$ on absorption and $2.0 \cdot 10^{-13}$ on fission — double precision round-off only.

Throughput on RTX 3080. On a 10^6-energy log-spaced grid inside U-238's RRR:

Implementation	ns/lookup	Relative
CPU single-thread (Ryzen 9800X3D)	66.8	1.00×
GPU (RTX 3080, saturated)	9.5	0.14×

7.0× per-lookup GPU-vs-CPU.

End-to-end GPU hybrid transport. The validated evaluator is wired into the event-based `transport_persistent` kernel: for each nuclide that carries WMP data, a bitmap `has_wmp[i]` and the corresponding $E^{\mathrm{WMP}}_{\mathrm{min}}, E^{\mathrm{WMP}}_{\mathrm{max}}$ are checked inline; if the incident energy falls in the resolved-resonance window, the SVD / pointwise lookup for elastic, fission and capture is replaced by `wmp_eval` at the nuclide's target temperature, the total XS is recomputed from partials, and URR sampling is suppressed (resonances are already explicit through the poles). Inelastic, $(n, 2n)$ and $(n, 3n)$ remain on the SVD path because WMP does not cover those channels.

To keep this practical on GPU the recursive conjugate fold in the Humlicek Faddeeva implementation is unrolled into a flag-plus-sign-flip form, because recursive `__device__` calls spill to local memory under the register pressure of `__launch_bounds__(256, 2)` and trigger CUDA_ERROR_ILLEGAL_ADDRESS on deep stacks. The rewritten evaluator is still bit-exact against the CPU reference at $5 \cdot 10^{-14}$.

On the PWR pin cell, 2 of 9 nuclides are covered by WMP (U-235 and U-238 in the fuel; the released ENDF/B-VII.1 WMP library does not ship U-234, O-16, H-1 or Zircaloy isotopes in a form the reader accepts). For those two nuclides the GPU hybrid path exercises `wmp_eval` throughout each nuclide's resolved-resonance window and falls back to the SVD / pointwise path elsewhere; covered nuclides therefore receive the explicit multipole treatment inside the RRR while uncovered ones keep the existing high-fidelity SVD path. k_∞ and throughput numbers from the large-L3 benchmark are in Section 6.

11 Threats to validity

Scope of benchmarks. Nine nuclides (thermal PWR pin cell) and three nuclides (fast Godiva). Full-core LWR analyses carry $\gtrsim 200$ nuclides; the hybrid memory picture scales with the reaction-channel mix and could look very different there.

S(α,β) coverage. Only H in H_2O is exercised. D_2O, graphite, ZrH, and other thermal libraries are not validated.

Sampling convention. Our original correlated-CDF sampling is a valid choice; we switched to OpenMC's stochastic-bin convention because that is what we were comparing against. A fully independent reference (MCNP, Serpent) would reveal whether OpenMC itself is biased, which we do not investigate here.

Charged-particle absorption. Channels MT= 103, 107, first-chance fission (MT= 19–21) are absorbed into the capture macroscopic. k_∞ is correct under this convention; MT-level reaction-rate tallies are not.

Statistical confidence. All comparisons use SEM $= \sigma_{\mathrm{seed}}/\sqrt{N_{\mathrm{seeds}}}$, not σ_{seed} itself. Large-L3 sweeps use $N_{\mathrm{seeds}} = 10$; hybrid sweep uses five.

Hybrid memory. The 132.9× representation-byte ratio is not an in-engine reduction. In-engine reduction is 1.06× (Sec. 9.4). The 132.9× density ratio is reported for what it is; the 1.06× engine-level reduction is the number an engine builder would care about.

GPU SVD. Slower than GPU pointwise on PWR pin cell because of the Zr clad level-sum cost. A per-warp level-sum cache would likely fix this but is not implemented.

GPU hybrid. Only the WMP evaluator itself is validated on GPU; the transport-loop integration is not done. No in-engine GPU hybrid k_∞ or throughput is reported.

12 Related work

Adaptive resampling machine-learned models [3, 4] retain 10–50% of the pointwise data with $\leq$ 97 pcm criticality error. Our approach is orthogonal (replace the representation, not subsample the array). Windowed Multipole for Doppler broadening is due to [5, 6]; hybrid multipole + windowed networks in RMC code [7] pair multipole with a second analytical method; our hybrid pairs multipole with SVD. Event-based GPU transport follows the Tramm architecture [8] and shares the memory-bottleneck diagnosis of earlier XSBench studies [9]. Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector (subspace angle $>$ $85°$ at $k = 2$ between ^{235}U, ^{238}U, ^{239}Pu), a finding that echoes KV-cache compression experience in a different domain [13].

13 Conclusion

Truncated SVD of the $\log_{10}$-transformed cross-section matrix is a defensible drop-in *representation* for the pointwise table. On a nine-nuclide thermal PWR pin cell, the same engine running rank-5 SVD vs its own pointwise table agrees on k_∞ within 13–47 pcm on two hardware tiers, and all four in-engine providers (CPU table, CPU SVD, GPU pointwise, GPU SVD) agree with OpenMC 0.15.3 to $|\Delta| \leq 51$ pcm at ten-seed statistics after a transport-level bug audit closed an initial 78–127 pcm engine offset. A Humlicek-W4 Faddeeva evaluator, pure-Rust WMP reader, and SVD+WMP hybrid provider all work end-to-end and reproduce k_∞ within two SEM of the matched SVD baseline at the same software revision; a CUDA port of the WMP evaluator is bit-exact against the CPU path and 7× faster per lookup on a consumer GPU.

On Godiva (HEU-MET-FAST-001) the four providers cluster 325–362 pcm *above* the ICSBEP experimental value 1.0000 ± 0.0010 [12]. That offset is the well-known ENDF/B-VII.1 Godiva bias — library re-runs at the same nuclide data systematically over-predict k_{eff} by 300–500 pcm, closing to $\sim$50 pcm under ENDF/B-VIII.0 — so we read this as our engine faithfully reproducing the library and not adding an engine-level offset on top of it. The four providers agree with each other on Godiva to within 37 pcm, consistent with the sub-SEM provider spread seen on PWR.

At a realistic *off-library* operating temperature, the story is more interesting. With a uniform +150 K offset applied to every nuclide (fuel 900 → 1050 K, moderator 600 → 750 K), OpenMC-style stochastic T-picking enabled on the pointwise and hybrid paths, and a 3-point unity Ducru weight scheme on the SVD path (Sec. 2.4), SVD reproduces k_∞ within 213 pcm of ACE+WMP at ten-seed 50 000-particle 120-batch statistics — a small ($\sim 3.6\sigma$) but real systematic, inside the publishable band. SVD is 22% smaller than WMP in memory and matches its speed within 0.07%. This is the result the paper exists to report: at a realistic operating temperature a compressed SVD representation of ENDF/B-VII.1 cross sections beats the industry ACE+WMP baseline on memory, ties it on speed, and trails it by a characterisable small systematic on k_∞ (Sec. 7.3).

Getting there was non-trivial. A naive 2-point unity Ducru leaves a 2094 pcm kernel-approximation bias at the same operating conditions (Sec. 7.2); a simplex-projected QP that clips the signed Ducru weights to non-negativity is *worse* (5% L2 error on U-238 capture versus 0.84% for the signed 3-point scheme) because the simplex projection is Euclidean-optimal in weight space whereas the raw Ducru weights are Euclidean-optimal in the Doppler-kernel norm

(Sec. 7.5). We also found and fixed a channel-consistency bug in our stochastic T-pick: drawing a fresh ξ per partial channel produced non-monotonic k_∞-versus-T behaviour that on-library measurements cannot detect (Sec. 7.1). The PHYSOR-style off-library test surfaced it as a byproduct of stressing the representation.

The *performance* story on-library is less one-sided. CPU SVD rank-5 is 1.37×–1.90× faster than the in-engine pointwise table on PWR pin cell (hardware-dependent) and 1.43× faster than the table on Godiva on the large-L3 system; GPU SVD is 1.30× *slower* than GPU pointwise on PWR because four clad nuclides require summing 13 discrete-level SVD kernels per lookup. The hybrid is 2.06× slower than CPU SVD because WMP evaluation is ∼15× more expensive than the SVD FMA per lookup.

The *memory* story is a correction of our earlier messaging. The representation-byte ratio of 132.9× (WMP + smooth-SVD vs four-channel six-temperature pointwise) is real; the in-engine reduction we measured is 1.06×, and the hybrid engine is absolutely 4.8–5.1× larger than the pointwise-table baseline. The discrete-level SVD kernels, which WMP does not cover, dominate the engine's working set. Reducing those kernels from rank 5 to rank 1 — the reviewer's suggestion and a free change because levels are weakly T-dependent (Sec. 7.4) — closes ≈70% of the engine's SVD working set on Godiva with no measurable k_{eff} shift. This is what enables the off-library 22% memory win against ACE+WMP referenced above: with discrete-level rank 1 applied, the SVD engine is absolutely smaller than the two-endpoint stochastic-T pointwise table even before the WMP comparison.

The practical takeaway for a reader considering this approach: rank-5 SVD with 3-point unity Ducru temperature interpolation and discrete-level rank-1 is correct, agrees with OpenMC and its own pointwise baseline within $\leq 3\,\text{SEM}_{10}$ on PWR at on-library T, and agrees with ACE+WMP within $\sim 3.6\,\text{SEM}_{10}$ on PWR at off-library T. On memory it wins modestly on-library and by 22% off-library; on speed it is 1.4×–1.9× faster than pointwise on CPU and at parity with ACE+WMP off-library. The 43/47 rank-one reactions and the weakly-T-dependent discrete levels are real structural properties of ENDF/B-VII.1 cross sections, worth exploiting; the raw Ducru weights are *not* a partition of unity and must be normalized before being applied to a $\log\sigma$ SVD.

We implemented and measured the kernel-Gram QP from [1] §4 that we initially flagged as future work (Sec. 7.5) and found a structural subtlety worth reporting: at low rank it beats the signed unity scheme (U-238 MT= 102 6.67 eV peak ratio 1.000 at rank 3 vs 0.994 for signed unity), but at our production rank 5 it is *worse* (0.983 vs 0.990 on the same peak). The QP's constrained optimum lies on the boundary of the probability simplex — one weight clips to zero, collapsing the 3-point problem back to a 2-point one — and at higher rank the signed unity scheme's negative weight on the far-away library column is actively useful as a shape-cancellation term. Non-negativity is the wrong constraint once basis-truncation error stops dominating. This is the single highest-leverage piece of follow-up work we did not publish in the original version of this paper; the result is that signed 3-point unity *is* the right production default at rank 5, not merely a convenient approximation to the QP ideal. We report the positive, mixed, and negative findings together so the trade-off is visible.

Data availability

The transport engine (pure Rust), the CUDA event-based solver, the CUDA WMP evaluator, all input decks, per-seed benchmark CSV files, the XS-audit scripts, the hybrid experiment Python script, and the scripts that generate the figures in this paper are released with the preprint. ENDF/B-VII.1 neutron and thermal data are available from the OpenMC data archive; the ENDF/B-VII.1 Windowed Multipole library is the public MIT release.

Declaration of AI assistance

This work was written with assistance from large language models for code and prose drafting; all physics, numerical experiments, and results were reviewed and validated by the human author.

References

[1] Pablo Ducru, Colin Josey, Kari Dibert, Vladimir Sobes, Benoit Forget, and Kord Smith. Kernel reconstruction methods for Doppler broadening — Temperature interpolation by linear combination of reference cross sections at optimally chosen temperatures. *Journal of Computational Physics*, 335:535–557, 2017. doi: 10.1016/j.jcp.2017.01.039.

[2] Paul K. Romano, Nicholas E. Horelik, Bryan R. Herman, Adam G. Nelson, Benoit Forget, and Kord Smith. OpenMC: A State-of-the-Art Monte Carlo Code for Research and Development. *Annals of Nuclear Energy*, 82:90–97, 2015. doi: 10.1016/j.anucene.2014.07.048.

[3] A. Hashemi, R. Macián-Juan, and M. Ohlerich. Development of a novel machine learning-based adaptive resampling algorithm for nuclear data processing. *Scientific Reports*, 15: 32573, 2025.

[4] A. Hashemi, R. Macián-Juan, and M. Ohlerich. Evaluating machine learned nuclear data precision in full core nuclear reactor Monte Carlo neutronics and computational efficiency analyses. *Scientific Reports*, 16:1314, 2026.

[5] Colin Josey, Pablo Ducru, Benoit Forget, and Kord Smith. Windowed multipole for cross section Doppler broadening. *Journal of Computational Physics*, 307:715–727, 2016. doi: 10.1016/j.jcp.2015.08.013.

[6] Benoit Forget, Sheng Xu, and Kord Smith. Direct Doppler broadening in Monte Carlo simulations using the multipole representation. *Annals of Nuclear Energy*, 64:78–85, 2014. doi: 10.1016/j.anucene.2013.09.043.

[7] Teng-Yue Huang, Zhi-Gang Li, Kan Wang, Xiao-Yu Guo, and Jin-Gang Liang. Hybrid windowed networks for on-the-fly Doppler broadening in RMC code. *Nuclear Science and Techniques*, 32(6):62, 2021.

[8] John Tramm, Paul Romano, Patrick Shriwise, Amanda Lund, Johannes Doerfert, Peter Steinbrecher, Andrew Siegel, and Gavin Ridley. Performance portable Monte Carlo particle transport on Intel, NVIDIA, and AMD GPUs. *EPJ Web of Conferences*, 302:04010, 2024.

[9] John R. Tramm and Andrew R. Siegel. Memory bottlenecks and memory contention in multi-core Monte Carlo transport codes. *Annals of Nuclear Energy*, 82:195–202, 2015. doi: 10.1016/j.anucene.2014.09.028.

[10] Forrest B. Brown. New hash-based energy lookup algorithm for Monte Carlo codes. *Transactions of the American Nuclear Society*, 111:659–662, 2014.

[11] John Duchi, Shai Shalev-Shwartz, Yoram Singer, and Tushar Chandra. Efficient projections onto the ℓ_1-ball for learning in high dimensions. In *Proceedings of the 25th International Conference on Machine Learning (ICML)*, pages 272–279, 2008. doi: 10.1145/1390156.1390191. Closed-form $O(n \log n)$ Euclidean projection of a vector onto the probability simplex; used here as a cheap approximation to the Ducru kernel-Gram QP of [1].

[12] Nuclear Energy Agency. International handbook of evaluated criticality safety benchmark experiments (icsbep), heu-met-fast-001: Bare HEU metal sphere (Godiva). Technical

Report NEA/NSC/DOC(95)03, OECD Nuclear Energy Agency, 2023. Benchmark value $k_{\text{eff}} = 1.0000 \pm 0.0010$.

[13] Akide Liu, Jing Zhao, et al. MiniCache: KV cache compression in depth dimension for large language models. In *Proceedings of NeurIPS*, 2024.